ACADEMIC PRESENTATION / CURATED PUBLIC EXCERPT

Crystal Defects and Dislocations

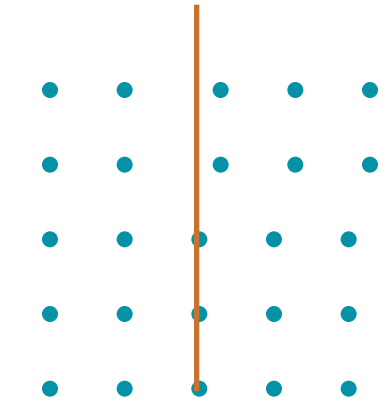
Semiconductor Technology / Academic presentation / 2025

SCIENTIFIC QUESTION

How do defects and dislocation motion affect crystalline materials?

KEY POINTS

- Point, line, surface and volume defects differ in dimension.
- Burgers vectors describe dislocation geometry and character.
- Glide, climb and multiplication affect crystal deformation.



MY CONTRIBUTION

Researched and presented defect classes, FCC/HCP examples, dislocation motion, jogs and Frank-Read multiplication.

LIMITS & ATTRIBUTION

Scientific literature synthesis; no sample fabrication, microscopy experiment or personally run simulation.

D. Hull and D. J. Bacon, Introduction to Dislocations.

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